Unit: IV) PERT and CPM

• Introduction:-

• PERT- Program Evaluation and Review Technique is a project management tool used to plan, schedule, and control complex Project.

In a PERT, a project divided into small tasks called activities.
Then we - a) Estimate the time required for each activity.
b) Decide the sequence (order) of activities.
c) calculate the total project time.

• CPM - Critical path method is a project management technique used to plan, schedule and control projects where the activity times are known and fixed.

In a CPM, a project ~~the~~ is divided into activities (task)
Then we - a) Assign a fixed time to each activity.
b) Decide the sequence (order) of activities.
c) Draw a network Diagram.
d) find the critical path.

* Basic Difference between PERT and CPM :-

Sr.No	PERT	CPM
1.	Project Evaluation and Review Technique.	Critical Path Method.
2.	PERT is probabilistic in nature and makes use of three completion time estimates of activities.	CPM is deterministic in nature and makes use of a single fixed time estimate for each activity.
3.	It is used for project involving non-repetative activities such as research, projects, new product.	It is used for project consisting of repetative activities such as construction of buiding.
4.	In PERT, there is more emphasis on the completion of task or event. It prepares the network from the event. So, it is event oriented technique.	CPM uses the activities to devlop the network. So, it is activity oriented techinque.
5.	PERT makes use only a time calculation in analysis.	CPM establishes time cost trade-offs for a project

* Phases of Project Management :-

1) Project Planning Phase :- This is the 1st step, where we plan everything before starting the project.

a) Identify all activities (what works need to be done.)

b) Decide resources needed (men, machines, materials, money)

c) Assign responsibilities to team members.

d) Allocate resources properly.

e) Estimate time and cost for each task.

f) Set performance standard.

g) Establish communication and control system.

2) Scheduling Phase :-

Here, we decide when each task will be done.

a) Identify people responsible for each task.

b) Estimate duration.

c) Find relationships between activities.

e) Draw a network Diagram.

f) Calculate total project time.

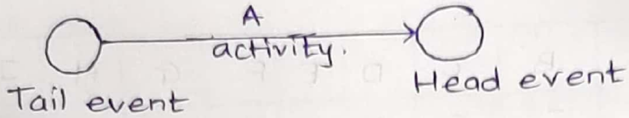
g) Identify Critical path.

3) Project Control phase :-

* Basic Terminology :-

1) **Activity** :- An activity means a piece of a work that takes time and efforts.

It is shown by an arrow (⟶) in a network Diagram.



2) **Event (Node)** :- An event is a point in time (not work).

It is represented by a circle (○).

3) **Tail event and Head event** :-

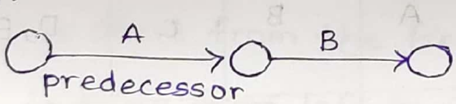
Tail event - Starting point of activity.

Head event - Ending point of activity.

- Activity always goes from tail → head.

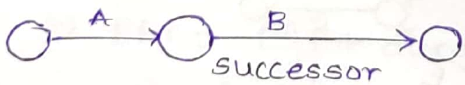
4) **Predecessor Activity** :-

Activity that must be completed before another activity starts.



5) **Successor Activity** :-

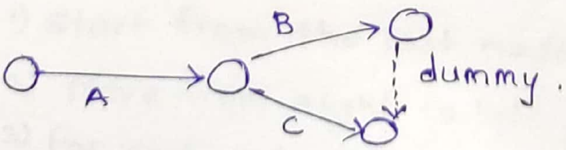
Activity that comes after another activity.



6) **Dummy Activity** :-

Activity with no time and no cost, used only to show ~~relash~~ relationships between activities.

Represented by dotted arrows.

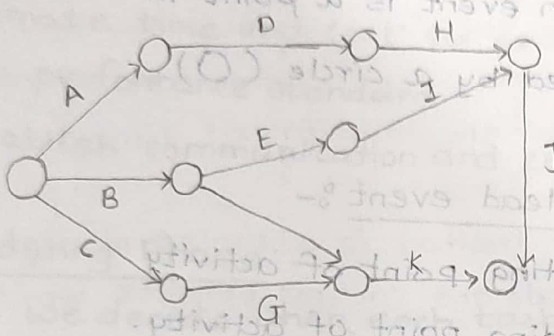


Network Diagram :-

1) Construct a Network Diagram for each of the project whose activities and their precedence relationships are given below.

Activity	A	B	C	D	E	F	G	H	I	J	K
Predecessor	–	–	–	A	B	B	C	D	E	H, I	F, G.

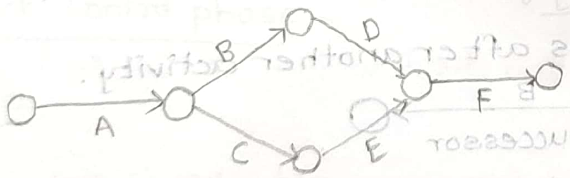
Soln.-



2) Draw a network Diagram.

Activity	A	B	C	D	E	F
predecessor Activity	–	A	A	B	C	D, E.

Soln:-



* Time Estimates :-

After drawing the network diagram, we calculate the time required to complete each activity and the whole project.

• Note :- 1) CPM ~~used~~ uses fixed time for each activity.

2) PERT uses three time estimates.

a) Optimistic time (O)

b) Most likely time (M)

c) Pessimistic time (P).

• Important ~~time :~~ terms :-

1) EST - Earliest start time.

2) EFT - Earliest finish time.

3) LST - Latest start time

4) LFT - Latest finish time.

• Forward Pass Computation :-

* Steps :- 1) start from the first node (time = 0)

2) Move from left to right.

3) For each activity,

a) EST = time of previous node

b) EFT = EST + duration.

4) If more than one path arrives at node then take the maximum value.

• Backward Pass Computation :-

* Steps :- 1) Start from the last node (TE).

2) Move from right to left.

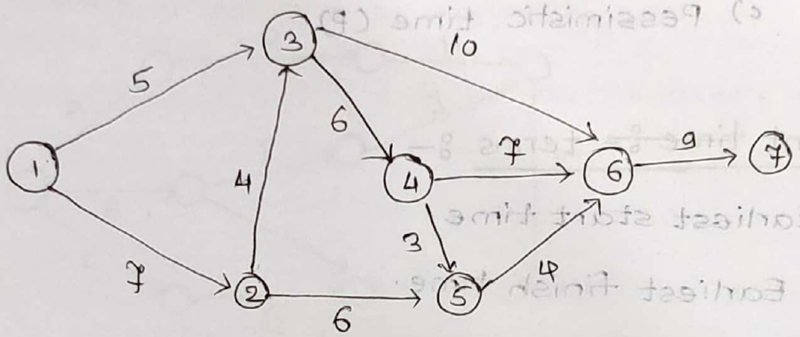
3) For each activity,

a) LFT = time to next node

b) LST = LFT - duration

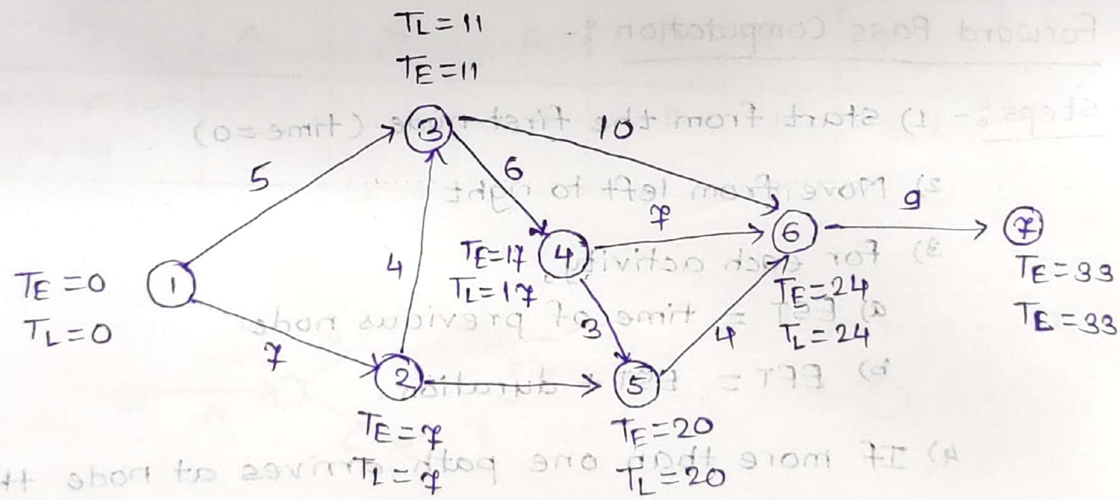
4) If more than one path leaves a node, then take the minimum value.

Que. Find the forward and backward pass computation for the following network Diagram. Also prepare the table showing EST, EFT, LST, LFT.



Sol^n:- Let, T_E - EST for forward pass computation.

T_L - LFT for backward pass computation.



Forward Pass Computation:

$T_E^{(1)} = 0$

$T_E^{(2)} = 7$

$T_E^{(3)} = \max\{5, 7+4\} = 11$

$T_E^{(4)} = \max\{5+6, 7+4+6\} = 17$

$$T_E^{(5)} = \max\{7+6, 17+3\} = 20$$

$$T_E^{(6)} = \max\{11+10, 17+7, 20+4\} = 24$$

$$T_E^{(7)} = 24+9 = 33.$$

Backward Pass Computation,

$$T_L^{(7)} = T_E^{(7)} = 33$$

$$T_L^{(6)} = T_E \; 33-9 = 24$$

$$T_L^{(5)} = 24-4 = 20$$

$$T_L^{(4)} = \min\{20-3, 24-7\} = 17.$$

$$T_L^{(3)} = \min\{24-10, 17-6\} = 11$$

$$T_L^{(2)} = \min\{20-6, 11-4\} = 7.$$

$$T_L^{(1)} = \min\{11-5, 7-7\} = 0$$

We prepare the table for EST, EFT, LST and LFT.

Activity	Duration	EST	EFT	LST	LFT
1-2	7	0	0+7=7	7-7=0	7
1-3	5	0	0+5=5	11-5=6	11
2-3	4	7	4+7=11	11-4=7	11
2-5	6	7	7+6=13	20-6=14	20
3-4	6	11	11+6=17	17-6=11	17
3-6	10	11	11+10=21	24-10=14	24
4-5	3	17	17+3=20	20-3=17	20
4-6	7	17	17+7=24	24-7=17	24
5-6	4	20	20+4=24	24-4=20	24
6-7	9	24	24+9=33	33-9=24	33

Critical Path and Various Floats for Activities :-

* Critical Activity :-

Certain Activities in a project network are called critical activities, because any delay in their execution causes delay in the completion of entire project

* Critical Path :-

Critical path is a sequence of the critical activities which starts from the initial event / node and goes upto terminal (last) ~~node~~ event / node through the network.

- Note :- The length of the critical path is the sum of duration of all the critical activities lies in the critical path.

* Event slack :-

It is the difference between latest finish time and earlist start time of that event.

for the ith event, $S^{(i)} = T_L^{(i)} - T_E^{(i)}$

* Activity float :-

The float of an activity is the available time by which its duration can be delayed.

There are three types of float

1) Total float (TF) :-

$$\underset{(TF)}{\text{Total float}} = LST - EST$$

- Note :- 1) For critical activity, the total float is zero.

2) Negative value of the Float implies that the activities may not be finished.

3) Total float is most important float because it gives the completion of the entire project

2) Free float (FF) :-

Free float (FF) = Total float - Head slack.

- Note :- Free float of an activity can have value from zero upto that of total float, but it can never be greater than the total float.

3) Independent float (IF) :-

Independent float (IF) = Free float - Tail slack.

- Note :- Independent float of an activity can has value at the most equal to free float but it can never be greater than it.

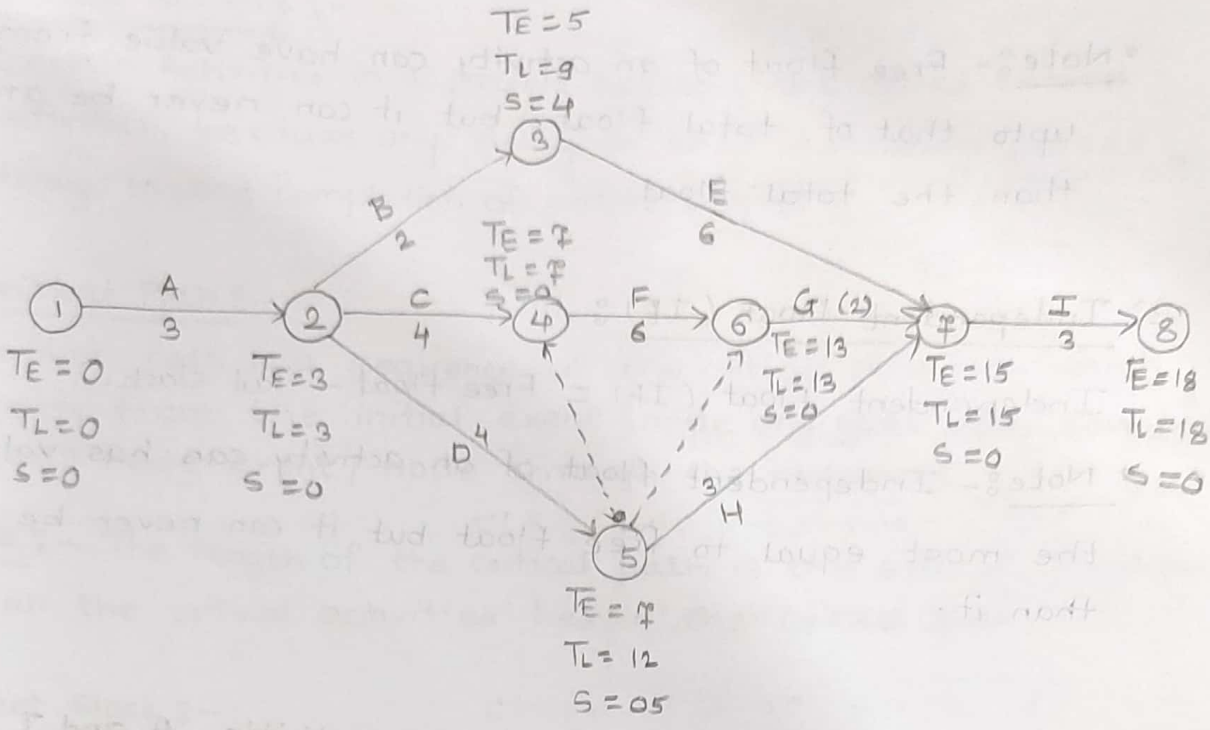
Q. The data regarding a project with activities. A and I are given below.

Activity	Preceding activity	Time (week)
A	-	3
B	A	2
C	A	4
D	A	4
E	B	6
F	C, D	6
G	D, F	2
H	D	3
I	E, G, H.	3

i) Draw the project network and find critical path.

ii) find total, free and independent floats for each activity.

Soln:-



We prepare activity Schedule Table.

Activity	Duration (weeks)	EST	EFT	LST	LFT	Total float	Head slack	Free float	Tail slack	Independent float
A = 1-2	3	0	3	0	3	0	0	0	0	0
2-3	2	3	5	7	9	4	4	0	0	0
2-4	4	3	7	3	7	0	0	0	0	0
2-5	4	3	7	8	12	5	5	0	0	0
3-7	6	5	11	9	15	4	0	4	4	0
4-6	6	7	13	7	13	0	0	0	0	0
6-7	2	13	15	13	15	0	0	0	0	0
5-7	3	7	10	12	15	5	0	5	5	0
7-8	3	15	18	15	18	0	0	0	0	0

Project Evaluation and Review Technique (PERT) :-

* **PERT :-** PERT is a project management technique used to plan and control projects when activity time is uncertain.

  - It is mainly used when -
    a) The project is new.
    b) Exact time duration are not known.
    c) There is uncertainty in completion time.

In PERT model requires three type time estimate for each activity.

* **Different types of time estimates :-**

1) **Optimistic time Estimates (t_o) :-**

It is the shortest possible time in which an activity can be performed assuming that everything goes well.

It is denoted by t_o.

2) ~~Pesimi~~ **Pessimistic time Estimate :- (t_p)**

It is the longest possible time required to perform an activity under extremely bad condition.

It is denoted by t_p.

3) **Most likely Time Estimate (t_m) :-**

It is the time that would occur most ~~opte~~ often to complete an activity.

It is denoted by t_m.

- **Note :-** $t_o \leq t_m \leq t_p$.

- Expected time of an activity (t_E) $= \dfrac{t_o + 4t_m + t_p}{6}$

- Variance of an activity time $\sigma^2 = \left(\dfrac{t_p - t_o}{6}\right)^2$.

- S.D. $= \sqrt{\left(\dfrac{t_p - t_o}{6}\right)^2}$

Calculation of PERT Network :-

• Steps :-

1) Compute expected duration and variance for each activity of the project.
2) Draw a project network based upon expected duration of the activity.
3) Find out the total expected duration of the project.
4) Find out the critical activities and critical path.
5) Find out S.D. of the project.

Q. A project has the following activities and characteristics.

Activity	Optimistic t_o	Most likely t_m	Pessimistic t_p.
1-2	2	5	8
1-3	4	10	16
1-4	1	7	13
2-5	8	8	11
3-5	2	8	14
4-6	6	9	12
5-6	4	7	10

Find out - 1) Expected duration for each activity.

2) Find varience of each activity

3) find varience of activities of critical path and its S.D.

Soln:- Let, t_o = Optimistic time estimates.

t_m = Most likely time estimates

t_p = Pessimistic time estimates in days.

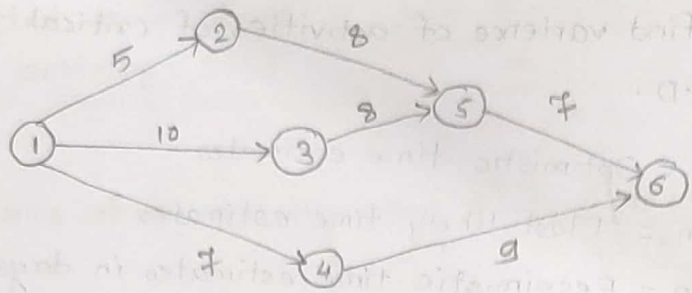
Now, to find,

Expected time $t_E = \dfrac{t_o + 4t_m + t_p}{6}$

and, variance $\sigma^2 = \left(\dfrac{t_p - t_o}{6}\right)^2$

Activity	Expected duration	Variance.
1-2	$\dfrac{2+(4\times5)+8}{6} = 5$	$\left(\dfrac{8-2}{6}\right)^2 = 1$
1-3	$\dfrac{4+(4\times10)+16}{6} = 10$	$\left(\dfrac{16-4}{6}\right)^2 = 4$
1-4	$\dfrac{1+(4\times7)+13}{6} = 7$	$\left(\dfrac{13-1}{6}\right)^2 = 4$
2-5	$\dfrac{5+(4\times8)+11}{6} = 8$	$\left(\dfrac{11-5}{6}\right)^2 = 1$
3-5	$\dfrac{2+(4\times8)+14}{6} = 8$	$\left(\dfrac{14-2}{6}\right)^2 = 4$
4-6	$\dfrac{6+(4\times9)+12}{6} = 9$	$\left(\dfrac{12-6}{6}\right)^2 = 1$
5-6	$\dfrac{4+(4\times7)+10}{6} = 7$	$\left(\dfrac{10-4}{6}\right)^2 = 1$

The network diagram for the given project is,



Various paths	Duration
1-2-5-6	5+8+7 = 20
1-3-5-6	10+8+7 = 25
1-4-6	7+9 = 16

Critical path = 1-3-5-6

Variance of critical activities = 4+4+1 = 9.

S.D. = $\sqrt{\text{varience}}$ = $\sqrt{9}$ = 3.

Q. The following table shows jobs of a network along with their time estimates.

Jobs	1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8
t_o	1	2	2	2	7	5	5	3	8
t_m	7	5	14	5	10	5	8	3	17
t_p	13	14	26	8	19	17	29	9	32

Draw the network diagram and find the probability that the project is completed in 40 days.

Soln:- Let, t_o - optimistic time

t_m - Most likely time

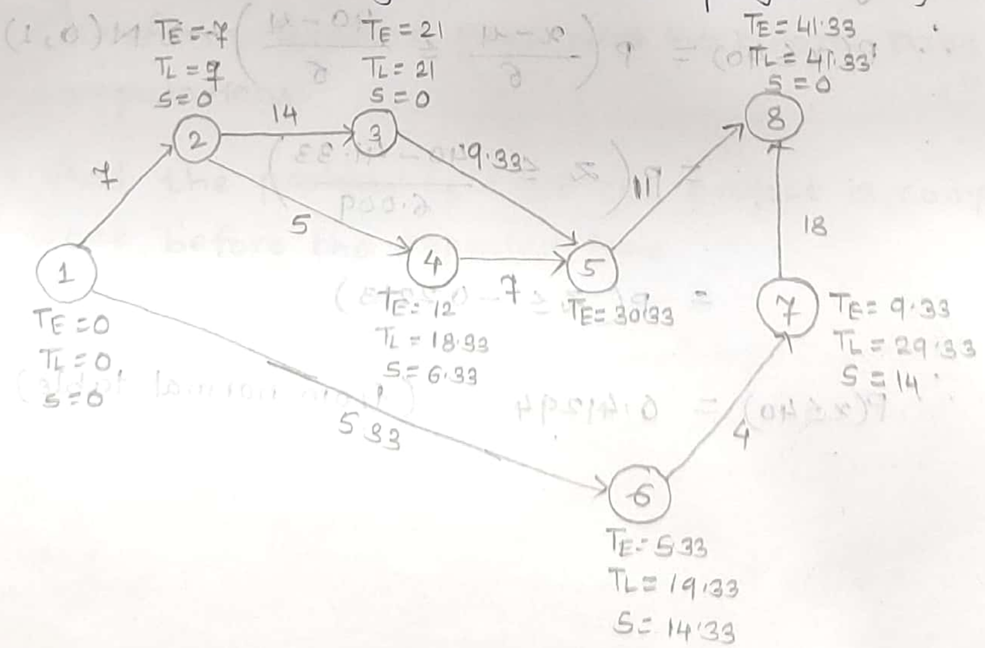
t_p - Pessimistic time estimates in days.

So, we find the variance and expected time for each activity.

Expected time $t_E = \frac{t_o + 4t_m + t_p}{6}$

Activity	Expected time	Variance
1-2	7	4
1-6	5.33	4
2-3	14	16
2-4	5	1
3-5	9.33	0.11
4-5	7	4
5-8	11	16
6-7	4	1
7-8	18	16

The Network diagram for the project is,



After finding forward pass and backward pass computation, we find the critical path.

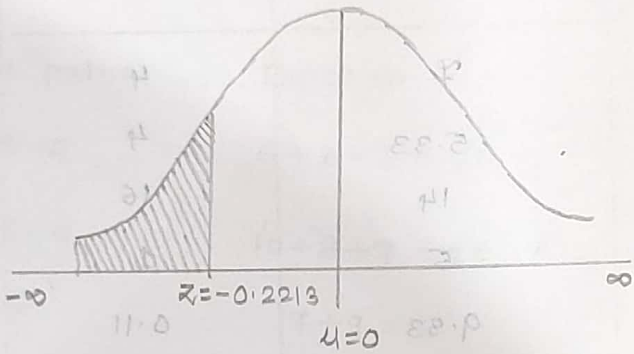
Critical path : 1 - 2 - 3 - 5 - 8

Project completed in : 41.33 days.

Variance of C.P. : $36.11 = \sigma^2$

Variance , $\sigma^2 = 36.11$

S.D. $\sigma = 6.009$.



Also, to find the probability that project is completed in 40 days.

$$P(x \leq 40) = ?$$

Using Normal approximation,

$$\therefore \ P(x \leq 40) = P\left(\frac{x-\mu}{\sigma} \leq \frac{40-\mu}{\sigma}\right) \rightarrow N(0,1)$$

$$= P\left(z \leq \frac{40 - 41.33}{6.009}\right)$$

$$= P(z \leq -0.2213)$$

$\therefore \ P(x \leq 40) = 0.41294.$ (from normal table).

Example for Practice :-

Q. A project consist of 9 activities whose time estimates in weeks and other characteristics are given below.

Activity	Preceding activity	most likely time (week)	Pessimistic time (week)	Optimistic time (week)
A	–	4	6	2
B	–	6	6	6
C	–	12	24	6
D	A	5	8	2
E	A	14	23	11
F	B, D	10	12	8
G	B, D	6	9	3
H	C, F	15	27	9
I	E	10	16	4

i) Draw the PERT network and find the critical path (perform forward pass and backward pass) computations.

ii) find the probability that the project is completed 1 week before the expected time.